3D Vectors & the Cross Product

Section 1: Dot Product and Orthogonality

1 Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = [2, -3, 5]$ and $\mathbf{v} = [4, 1, -2]$.

a. -5

b. 5

c. 15

d. -1

2 For what value of k are the vectors $\mathbf{a} = [k, 3, -2]$ and $\mathbf{b} = [4, -2, 5]$ orthogonal (perpendicular)?

a. 4

b. 2

c. -4

d. 5

3 Find the angle θ between $\mathbf{u} = [1, 1, 0]$ and $\mathbf{v} = [0, 1, 1]$.

a. $\frac{\pi}{3}$ (60°)

b. $\frac{\pi}{4}$ (45°)

c. $\frac{\pi}{6}$ (30°)

d. $\frac{\pi}{2}$ (90°)

Section 2: The Vector Cross Product Formula

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4 Compute the cross product $\mathbf{u} \times \mathbf{v}$ where $\mathbf{u} = [1, 2, 3]$ and $\mathbf{v} = [4, 5, 6]$.

a. $[-3, 6, -3]$

b. $[3, -6, 3]$

c. $[-3, -6, -3]$

d. $[7, -2, -3]$

5 What is the geometric relationship between the cross product vector $\mathbf{u} \times \mathbf{v}$ and the vectors $\mathbf{u}$ and $\mathbf{v}$?

a. $\mathbf{u} \times \mathbf{v}$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$

b. $\mathbf{u} \times \mathbf{v}$ is parallel to $\mathbf{u} + \mathbf{v}$

c. $\mathbf{u} \times \mathbf{v}$ bisects the angle between $\mathbf{u}$ and $\mathbf{v}$

d. $\mathbf{u} \times \mathbf{v}$ lies in the plane spanned by $\mathbf{u}$ and $\mathbf{v}$

6 If $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ for non-zero vectors $\mathbf{u}$ and $\mathbf{v}$, what does this indicate?

a. The vectors are parallel (collinear)

b. The vectors are perpendicular

c. The vectors have unit magnitude

d. The vectors are identical

Section 3: Area of Triangles and Parallelograms

7 Find the area of the parallelogram spanned by vectors $\mathbf{a} = [2, 0, 1]$ and $\mathbf{b} = [1, 3, -2]$.

a. $\sqrt{59}$

b. $\sqrt{35}$

c. 14

d. $\frac{\sqrt{59}}{2}$

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8 Find the area of the triangle with vertices $A(1,0,0), B(0,2,0), C(0,0,3)$.

a. $\frac{7}{2}$

b. 7

c. $\sqrt{14}$

d. 3

9 If two vectors have lengths $|\mathbf{u}| = 6$ and $|\mathbf{v}| = 5$, and the angle between them is 30°, what is $|\mathbf{u} \times \mathbf{v}|$?

a. 15

b. $15\sqrt{3}$

c. 30

d. $30\sqrt{3}$

Section 4: Scalar Triple Product & Volume

10 What geometric quantity is given by the absolute value of the scalar triple product $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$?

a. The volume of the parallelepiped formed by $\mathbf{a}, \mathbf{b}, \mathbf{c}$

b. The surface area of the prism formed by $\mathbf{a}, \mathbf{b}, \mathbf{c}$

c. The length of the main diagonal vector

d. The perimeter of the base

11 Calculate the volume of the parallelepiped with edges $\mathbf{a} = [1,0,0], \mathbf{b} = [0,2,0], \mathbf{c} = [1,1,3]$.

a. 6

b. 3

c. 12

d. 2

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12 If $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$, what does this reveal about the three vectors?

a. The three vectors are coplanar (lie in the same plane)

b. The three vectors are mutually perpendicular

c. At least one vector must be the zero vector

d. The vectors form an equilateral triangle

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